

Note: Use python tutor for tracing

Day 7–8: Loops

- for loop
- while loop
- do-while
- Nested loops

Programs:

- Patterns
- Prime number
- Fibonacci
- Factorial

Chapter 1: What is a Loop?

First oka real-life example chuddam.

Imagine...

Teacher chepparu:

Write "I Love Java"

100 times.

Without loops:

```
System.out.println("I Love Java");
```

```
System.out.println("I Love Java");
```

```
System.out.println("I Love Java");
```

...

100 times

😓 Very boring.

Instead Java chepthundi

Repeat this statement 100 times.

This repeating mechanism is called

LOOP

Definition:

Loop ante

same block of code ni multiple times execute cheyyadam.

Simple ga

Repeat

Repeat

Repeat

Repeat

until condition becomes false

Why Loops?

Without loops

Huge code

Time waste

More mistakes

Hard to maintain

With loops

Small code

Easy

Fast

Readable

Example

Without loop

```
System.out.println(1);
```

```
System.out.println(2);
```

```
System.out.println(3);
```

```
System.out.println(4);
```

```
System.out.println(5);
```

With loop

```
for(int i=1;i<=5;i++)
```

```
{
```

```
    System.out.println(i);
```

```
}
```

Same output.

Only 3 lines.

Types of Loops

Java has 3 main loops.

for loop

while loop

do-while loop

Plus

Nested loops

We'll learn one by one.

FOR LOOP

This is the most used loop in Java.

Syntax

```
for(initialization; condition; increment/decrement)
```

```
{  
    statements;  
}
```

Looks confusing?

Don't worry.

Let's divide.

```
for(int i=1;i<=5;i++)
```

This has 3 parts.

Part 1

Initialization

```
int i=1;
```

Meaning

Create variable

Start from 1

Runs only once.

Part 2

Condition

```
i<=5
```

Question Java asks every iteration.

Is $i \leq 5$?

YES

Execute

NO

Stop

Part 3

Increment

`i++`

After execution

Increase value.

`1→2→3→4→5→6`

Let's dry run.

```
for(int i=1;i<=5;i++)
```

```
{
```

```
    System.out.println(i);
```

```
}
```

Iteration 1

`i=1`

`1<=5`

true

print 1

`i++`

2

Iteration 2

`2<=5`

true

print 2

i++

3

Iteration 3

3

Iteration 4

4

Iteration 5

5

Iteration 6

6<=5

false

stop

Output

1

2

3

4

5

Flow Diagram

Start



Initialization



Condition



Statements



Increment



Condition



Statements



Increment

↓

False

↓

Stop

Infinite Loop

```
for(;;)
{
    System.out.println("Hello");
}
```

No condition.

Never stops.

Infinite loop.

Decrement Loop

```
for(int i=10;i>=1;i--)
```

Output

10

9

8

...

1

Skip Numbers

```
for(int i=2;i<=20;i=i+2)
```

Output

```
2
4
6
8
10
12
14
16
18
20
```

WHILE LOOP

Used when

We don't know exactly how many times loop should run.

Syntax

```
while(condition)
```

```
{
    statements;
}
```

Example

```
int i=1;
```

```
while(i<=5)
```

```
{
    System.out.println(i);
```

```
    i++;  
}
```

Dry run

i=1

1<=5

print

i++

2<=5

print

...

6<=5

false

stop

Difference from for loop

For loop

Everything in one line.

```
for(int i=1;i<=5;i++)
```

While loop

Initialization outside.

Increment inside.

```
int i=1;
```

```
while(i<=5)
```

```
{
```

```
    System.out.println(i);
```

```
    i++;
```

```
}
```

Common Mistake

```
int i=1;
```

```
while(i<=5)
```

```
{
```

```
    System.out.println(i);
```

```
}
```

Forgot

```
i++;
```

Output

1

1

1

1

1

1

1

Infinite loop.

DO WHILE LOOP

Special loop.

Difference:

Condition checked AFTER execution.

Syntax

do

{

 statements;

}

while(condition);

Example

int i=1;

do

{

 System.out.println(i);

 i++;

}

while(i<=5);

Output

1

2

3

4

5

Example

int i=10;

do

```
{  
    System.out.println(i);  
}
```

while(i<5);

Output

10

Why?

Because

do block executes first.

Then checks condition.

Difference

For

Condition first.

While

Condition first.

Do While

Execution first.

Comparison

Feature	for	while	do-while
Initialization	Inside	Outside	Outside
Condition	Before	Before	After
Executes at least once	No	No	Yes
Best use	Fixed iterations	Unknown iterations	Menu programs

NESTED LOOPS

Nested means

Loop inside another loop.

Example

```
for(int i=1;i<=3;i++)  
{  
    for(int j=1;j<=2;j++)  
    {  
        System.out.println(i+" "+j);  
    }  
}
```

Outer loop

i

Inner loop

j

Dry run

Outer

i=1

Inner

1 1

1 2

Outer becomes

2

Inner again starts from beginning

2 1

2 2

Outer

3

Inner

3 1

3 2

Output

1 1

1 2

2 1

2 2

3 1

3 2

Important Rule:

Outer loop oka sari run ayina prati sari, inner loop malli starting nundi complete ga run avutundi.

Pattern Printing

Pattern questions almost always use **nested loops**.

Example 1

*
*
*
*
*

```
for(int i=1;i<=5;i++)
```

```
{
```

```
    System.out.println("*");  
}
```

Example 2

```
for(int i=1;i<=5;i++)  
{  
    System.out.print("*");  
}
```

Difference:

println() → next line.

print() → same line.

Example 3

*

**

```
for(int i=1;i<=5;i++)  
{  
    for(int j=1;j<=i;j++)  
    {  
        System.out.print("*");  
    }  
    System.out.println();  
}
```

Dry run

i=1


```
*  
  
i=2  
  
**  
  
i=3  
  
***  
  
i=4  
  
****  
  
i=5  
  
*****
```

Example 4

```
*****  
  
****  
  
***  
  
**  
  
*  
  
for(int i=5;i>=1;i--)  
{  
    for(int j=1;j<=i;j++)  
    {  
        System.out.print("*");  
    }  
    System.out.println();  
}
```

Example 5

```
1  
  
12  
  
123
```

1234

12345

```
for(int i=1;i<=5;i++)
{
    for(int j=1;j<=i;j++)
    {
        System.out.print(j);
    }
    System.out.println();
}
```

Prime Number

Definition:

Prime number ante 1 kanna pedda number, and 1 and aa number thappa vere number tho divide avvakudadhu.

Examples:

2 ☒

3 ☒

5 ☒

7 ☒

11 ☒

Not Prime:

4

6

8

9

10

Prime numbers ki exactly **2 factors** untayi:

- 1
- The number itself

Prime check concept ni visual ga ardham chesukovadaniki:

xxx1004222555

100 factors into $2 \times 2 \times 5 \times 5$.

n

n

Logic

Count factors.

int n=7;

int count=0;

for(int i=1;i<=n;i++)

{

if(n%i==0)

{

count++;

}

}

if(count==2)

{

```
        System.out.println("Prime");
    }
    else
    {
        System.out.println("Not Prime");
    }
```

Dry run for $n = 7$

$7 \% 1 = 0$ ✓

$7 \% 2 \neq 0$

$7 \% 3 \neq 0$

$7 \% 4 \neq 0$

$7 \% 5 \neq 0$

$7 \% 6 \neq 0$

$7 \% 7 = 0$ ✓

Factors = 2 $\rightarrow$ Prime.

Fibonacci Series

Sequence:

0

1

1

2

3

5

8

13

21

34

...

Rule:

Next = Previous + Current

Example:

0+1=1

1+1=2

1+2=3

2+3=5

Program

```
int a=0;
```

```
int b=1;
```

```
System.out.print(a+" "+b+" ");
```

```
for(int i=3;i<=10;i++)
```

```
{
```

```
    int c=a+b;
```

```
    System.out.print(c+" ");
```

```
    a=b;
```

```
    b=c;
```

```
}
```

Output

0 1 1 2 3 5 8 13 21 34

Memory update:

Initially

$a=0$

$b=1$

$c=1$

↓

$a=1$

$b=1$

↓

$c=2$

↓

$a=1$

$b=2$

↓

c=3

...

Factorial

Definition:

Factorial ante 1 nundi aa number varaku anni numbers ni multiply cheyyadam.

Notation:

5!

Means

$5 \times 4 \times 3 \times 2 \times 1$

Answer

120

Examples

$3! = 6$

$4! = 24$

$5! = 120$

$6! = 720$

Program

```
int n=5;
```

```
int fact=1;
```

```
for(int i=1;i<=n;i++)
```

```
{
```

```
    fact=fact*i;
```

```
}
```

```
System.out.println(fact);
```

Dry run

fact=1

1×1=1

1×2=2

2×3=6

6×4=24

24×5=120

Output

120

Loop Control Statements (Bonus)

break

Loop ni immediate ga stop chestundi.

```
for(int i=1;i<=10;i++)
```

```
{
```

```
    if(i==5)
```

```
        break;
```

```
    System.out.println(i);
```

```
}
```

Output:

1

2

3

4

continue

Current iteration ni skip chesi next iteration ki velthundi.

```
for(int i=1;i<=5;i++)
```

```
{
```

```
    if(i==3)
```

```
        continue;
```

```
    System.out.println(i);
```

```
}
```

Output:

1

2

4

5

Real-Life Uses of Loops

-  ATM menu repeatedly show cheyyadam.
 -  Game lo player alive unna varaku game run avvadam.
 -  Mobile apps notifications check cheyyadam.
 -  Website lo comments list display cheyyadam.
 -  Excel rows process cheyyadam.
 -  AI training lo millions of data records iterate cheyyadam.
 -  Folder lo files read cheyyadam.
-

Common Beginner Mistakes

1. ✗ `i++` marchipovadam → Infinite loop.
 2. ✗ Wrong condition (`i < 5` vs `i <= 5`).
 3. ✗ `print()` badulu `println()` use cheyyadam pattern questions lo.
 4. ✗ Nested loop lo outer/inner variables confuse avvadam.
 5. ✗ do-while lo `while(condition);` tarvata semicolon `;` marchipovadam.
-

Interview Questions

1. What is a loop?
 2. Difference between for, while, and do-while.
 3. When should you use each loop?
 4. What is an infinite loop?
 5. What is a nested loop?
 6. Why are nested loops used for pattern printing?
 7. What is the difference between break and continue?
 8. Explain the execution flow of a for loop.
 9. How do you check whether a number is prime?
 10. Write logic for factorial and Fibonacci series.
-

Summary

- **Loop** → Repeats code multiple times.
- **for loop** → Best when the number of iterations is known.
- **while loop** → Best when iterations depend on a condition.
- **do-while loop** → Executes at least one time before checking the condition.
- **Nested loops** → Loop inside another loop, mainly used for patterns and matrix-like problems.
- **Prime** → Number greater than 1 with exactly 2 factors.
- **Fibonacci** → Every new number is the sum of the previous two.
- **Factorial** → Product of all numbers from 1 to n.

- **break** stops the loop immediately, while **continue** skips only the current iteration.